

## Lab 6

- In this lab it is required to implement an aggregation behavior
- The Implementation goes as follows, each robot has two states moving (0) in which it performs random walk and obstacle avoidance and stopped (1) in which the robot remains stationary
- Robots use the RAB (range-and-bearing) sensor to communicate.
  - Moving robots broadcast 0
  - Stopped robots broadcast 1
  - Each robot counts the number of nearby stopped robots within a certain range
- There are two probability rules
  - Stopping probability:

$$P_s = \min(P_s^{max}, S + \alpha N)$$

- Walking probability:

$$P_w = \max(P_w^{min}, W - \beta N)$$

- So a moving robot is more likely to stop when more robots nearby are stopped
- A stopped robot is less likely to leave when surrounded by many stopped robots
- Obstacle avoidance is implemented using proximity sensors and robots turn away from detected obstacles while moving
- When simulation starts the robots initially move around the arena, small groups begin to form as robots stop near other stopped robots
- Larger groups become more stable because robots inside the group rarely leave
- Robots at the edge of the group are more likely to resume moving